ASHA AI | EARLY HEALTH RISK PREDICTOR

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TEAM MEMBERS:

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PROBLEM STATEMENT

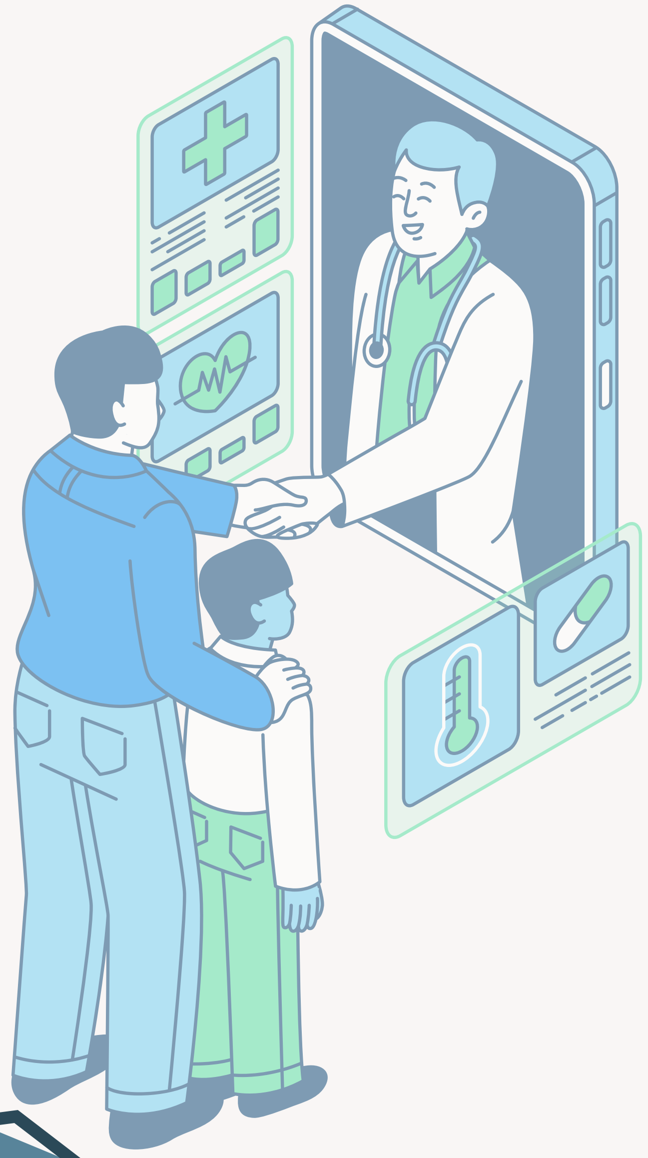
Early detection of health risks in rural populations is difficult due to limited hospital access, low health awareness, poor connectivity, and the absence of AI-driven predictive healthcare systems, despite the presence of ASHA health workers who currently lack intelligent digital tools for early risk identification.

ASHA workers - They are the triained nurse appoineted by trusts and governmenet to take analysys of heath data in rural areas

ISSUES IN CURRENT FLOW

- Health data collected by ASHA workers **is mostly paper-based or fragmented**
- Periodic data is not digitalized or analyzed over time
- No AI or predictive models to identify early health risks
- Diseases are detected only after symptoms become severe
- Limited access to doctors causes delayed medical intervention
- Poor network connectivity prevents real-time monitoring
- Preventive healthcare is largely absent in rural areas

PROPOSED SOLUTION & CURE



- Utilize ASHA health workers to periodically collect patient health data during routine visits using the asha mobile app we are building.
- Digitize collected data including vitals, symptoms, and basic health reports
- Apply AI/ML models to analyze trends and patterns in patient data
- Generate early health risk predictions for conditions such as diabetes, hypertension, anemia, and cardiac risks
- Provide risk alerts and recommendations to healthcare authorities for timely intervention
- Support offline data collection with synchronization when connectivity is available

PRODUCT DESCRIPTION

- A tablet(Android)-based healthcare application used by ASHA health workers in rural areas
- Enables digital patient registration and periodic health data collection
- Stores patient vitals, symptoms, and basic health observations securely
- Backend AI engine analyzes collected data to predict early health risks
- Generates risk scores and alerts for high-risk individuals
- Dashboard access for doctors and healthcare administrators
- Designed to work in low-connectivity rural environments
- Can be used to reduce risk of spread of disease and controll at the begining and can be used for pregnant womens and childern

TECH STACK & IMPLEMENTATION

- Data Collection: Android Tablet Application (used by ASHA health workers for periodic data collection)
- Frontend: Android Application (React pwa)
- Backend: Python (Flask / FastAPI)
- AI / ML: XGBoost – Health risk prediction on structured patient data
  LLaMA 3 – 8B Instruct – Explains prediction results
- Libraries: transformers, torch, accelerate, xgboost
- Language & Accessibility: Bhashini -Enables multilingual interaction
- Database: Secure Patient Health Records Database
- Cloud: AI model hosting, analytics, and centralized monitoring dashboard
- Connectivity: Offline-first system with periodic data synchronization for low-connectivity rural areas

KEY FEATURES & LIMITATIONS

- AI-based early health risk prediction
- Periodic health data collection using ASHA workers
- Multilanguage support
- chat - bot
- Risk scoring and trend analysis for patients
- Preventive alerts for high-risk individuals
- Digital health records for rural patients
- Works in low-connectivity environments and can update daily in their office when network available
- In the same application we can take care of pregnant womens periodic checkup and medical drops and vaccine like polio for childern

BUSINESS MODULE

Target Stakeholders

- Government healthcare departments
- NGOs working in rural healthcare

Value Proposition

- Early health risk prediction using AI
- Better decision-making using real-time health insights

Revenue / Sustainability Model

- Government-funded deployment under public healthcare schemes
- Institutional partnerships with NGOs and health organizations

Cost Structure

- Application development and maintenance
- Tablets and basic training for ASHA workers

Scalability

- Can be expanded to multiple villages and districts